

Count-Min Sketch

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Frequency Estimation with Deletions

- ▶ **Stream:** Sequence of updates (x, Δ) where x is an item and $\Delta \in \{+1, -1\}$.
- ▶ $+1$: insertion of item x ; -1 : deletion of item x .
- ▶ **Frequency:** $f_x = \sum_{(x, \Delta) \text{ in stream}} \Delta$.
- ▶ **Heavy Hitters Problem:** Given parameter k , find all items with $f_x > \|f\|_1/k$ where $\|f\|_1 = \sum_y |f_y|$.
- ▶ **Goal:** Return all items with $f_x > \|f\|_1/k$ and no items with $f_x < \|f\|_1/(2k)$.
- ▶ **Memory:** Sublinear in number of distinct items.

Data Structure

- ▶ d hash functions $h_1, \dots, h_d : \mathcal{U} \rightarrow \{1, \dots, w\}$.
- ▶ $d \times w$ array C of counters, initially zero.
- ▶ **Update**(x, Δ): For $i = 1$ to d : $C[i][h_i(x)] \leftarrow C[i][h_i(x)] + \Delta$.
- ▶ **Query**(x): $\tilde{f}_x = \min_{i=1}^d C[i][h_i(x)]$.

Visualization: Insertion

Insert $(x, +1)$: hash x in each row, **increment** the corresponding counter.

Before								After						
h_3	1	0	3	0	2	1	$\Rightarrow$	h_3	1	0	3	0	3	1
h_2	0	4	2	1	0	3		h_2	0	5	2	1	0	3
h_1	3	0	1	2	0	1		h_1	3	0	2	2	0	1

- ▶ $h_1(x) = 3$, $h_2(x) = 2$, $h_3(x) = 5$: each highlighted counter incremented by 1.
- ▶ Each item hashes to **one counter per row** (d rows total).

Visualization: Deletion

Delete $(x, -1)$: hash x in each row, **decrement** the corresponding counter.

Before								After						
h_3	1	0	3	0	3	1	$\Rightarrow$	h_3	1	0	3	0	2	1
h_2	0	5	2	1	0	3		h_2	0	4	2	1	0	3
h_1	3	0	2	2	0	1		h_1	3	0	1	2	0	1

- ▶ $h_1(x) = 3$, $h_2(x) = 2$, $h_3(x) = 5$: each highlighted counter decremented by 1.
- ▶ Counters may remain positive due to **collisions** with other items.
- ▶ Collisions cause **overestimation**; the min across rows mitigates this.

Visualization: Query

Query(x): hash x in each row, read counters, return the **minimum**.

h_3	0	0	1	0	3	0	$\rightarrow 3$
h_2	0	1	0	2	0	0	$\rightarrow 1$
h_1	0	0	2	0	0	1	$\rightarrow 2$

$$\tilde{f}_x = \min(3, 1, 2) = 1$$

- Read each **highlighted** counter; return the minimum.
- Rows with collisions overestimate; the min gives the tightest bound.
- Guarantee: $f_x \leq \tilde{f}_x \leq f_x + \varepsilon \|f\|_1$.

Goal: Heavy Hitters

Given parameter k :

- ▶ **Report** all items with $f_x > \|f\|_1/k$.
- ▶ **Don't report** any item with $f_x < \|f\|_1/(2k)$.

Strategy: Set $w = 2k$ columns and $d = O(\log n)$ rows.

Report item x if $\tilde{f}_x > \|f\|_1/(2k)$.

Estimate Never Underestimates

Claim: $\tilde{f}_x \geq f_x$ always.

Each counter accumulates f_x plus collisions from other items:

$$C[i][h_i(x)] = f_x + \underbrace{\sum_{y \neq x: h_i(y)=h_i(x)} f_y}_{\geq 0} \geq f_x$$

Error in a Single Row

Define the overestimation error in row i :

$$X_i = C[i][h_i(x)] - f_x = \sum_{y \neq x: h_i(y)=h_i(x)} f_y$$

By pairwise independence, each $y \neq x$ collides with probability $1/w$:

$$\mathbb{E}[X_i] = \sum_{y \neq x} f_y \cdot \frac{1}{w} \leq \frac{\|f\|_1}{w} = \frac{\|f\|_1}{2k}$$

Applying Markov's Inequality

We want: $X_i < \|f\|_1/(2k)$, i.e., overestimation is small.

By Markov's inequality on $X_i \geq 0$:

$$\Pr\left(X_i \geq \frac{\|f\|_1}{2k}\right) \leq \frac{\mathbb{E}[X_i]}{\|f\|_1/(2k)} \leq \frac{\|f\|_1/(2k)}{\|f\|_1/(2k)} = 1$$

This is too weak! We need a gap. Use $w = 2ek$ instead:

$$\mathbb{E}[X_i] \leq \frac{\|f\|_1}{2ek}, \quad \Pr\left(X_i \geq \frac{\|f\|_1}{2k}\right) \leq \frac{\|f\|_1/(2ek)}{\|f\|_1/(2k)} = \frac{1}{e}$$

Combining All Rows

The estimate is bad only if **every** row has large error.

Since hash functions are independent:

$$\Pr\left(\tilde{f}_x > f_x + \frac{\|f\|_1}{2k}\right) \leq \left(\frac{1}{e}\right)^d$$

With $d = \lceil \ln n \rceil$ (where n is the number of distinct items):

$$\left(\frac{1}{e}\right)^d \leq \frac{1}{n}$$

By a union bound over all n items, the sketch is correct for **every** item simultaneously with probability $\geq 1 - n/n = 0 \dots$ we need $d = \lceil 2 \ln n \rceil$ to get failure $\leq 1/n^2$, so union bound gives $\leq 1/n$.

Correctness of Heavy Hitters

With $w = 2ek$ and $d = \lceil 2 \ln n \rceil$, w.h.p. for all items x :

$$f_x \leq \tilde{f}_x \leq f_x + \frac{\|f\|_1}{2k}$$

Report x if $\tilde{f}_x > \|f\|_1/k$:

- ▶ **True heavy hitter** ($f_x > \|f\|_1/k$): $\tilde{f}_x \geq f_x > \|f\|_1/k$. Always reported.
- ▶ **Light item** ($f_x < \|f\|_1/(2k)$):

$$\tilde{f}_x \leq f_x + \frac{\|f\|_1}{2k} < \frac{\|f\|_1}{2k} + \frac{\|f\|_1}{2k} = \frac{\|f\|_1}{k}$$

So $\tilde{f}_x < \|f\|_1/k$. Never reported.

- ▶ Items with $f_x \in [\|f\|_1/(2k), \|f\|_1/k]$ may or may not be reported.

Space and Time

- ▶ **Space:** $d \times w = O(k \log n)$ counters.
- ▶ **Update time:** $O(d) = O(\log n)$ hash computations per update.
- ▶ **Query time:** $O(d) = O(\log n)$ hash computations + minimum.

Summary

- ▶ Count-Min Sketch finds heavy hitters using $O(k \log n)$ space.
- ▶ Reports all items with $f_x > \|f\|_1/k$.
- ▶ Never reports items with $f_x < \|f\|_1/(2k)$.
- ▶ Parameters: $w = O(k)$ columns, $d = O(\log n)$ rows.
- ▶ Extensions: Count Sketch (median instead of min), conservative update.